Division **E**

Mathematical Olympiads
November 2025
for Elementary & Middle Schools

Contest **1**

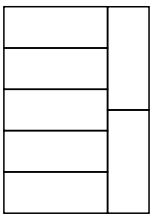
Directions to Students: After all questions have been read by your PICO, you will have 30 minutes to complete this contest. You may not have a pen or pencil in your hand while the PICO reads the set of questions to the class. Calculators are not permitted. Answers must be placed in the corresponding boxes in the answer column.

Name: _____

1A Find the following sum:

$$\begin{array}{r} 7 \\ 77 \\ 777 \\ 7777 \\ +\,77777 \\ \hline ? \end{array}$$

1B Seven congruent rectangles are arranged to form a large rectangle with area 70 square cm, as shown in the diagram. The dimensions of each small rectangle are whole numbers. What is the number of cm in the perimeter of the large given rectangle?



1C Ashley has twice as many stickers as Beth. Beth has 6 more stickers than Connie. Deb has triple the number of stickers that Connie has. Together, the four girls have 46 stickers. How many stickers does Beth have?

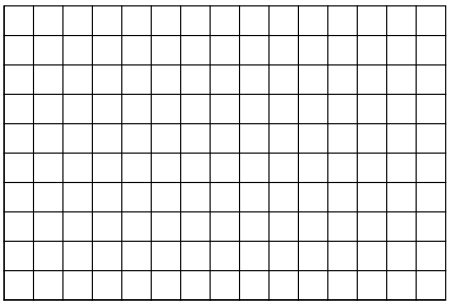
Answer Column
1A
1B
1C
1D
1E
Do Not Write in this Space. For PICO's Use Only. *SCORE:*

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Name: _____

1D A list of numbers begins with 20 and each following number is 24 more than the previous number: 20, 44, 68, 92, and so on. The last number is 1004. How many three-digit numbers are in the list?

1E In a grid of 1×1 squares, Melissa starts shading some 1×1 squares. She will not shade a square that would share anything, including a corner, with any other shaded 1×1 square. What is the maximum number of 1×1 squares that could be shaded in a 10×15 grid?



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SOLUTIONS AND ANSWERS

1A **METHOD 1** *Strategy: Use place value to efficiently add the numbers.*

There is one 7 in the ten-thousands place: $1 \times 7 \times 10{,}000 = 70{,}000$
There are two 7s in the thousands place: $2 \times 7 \times 1000 = 14{,}000$
There are three 7s in the hundreds place: $3 \times 7 \times 100 = 2{,}100$
There are four 7s in the tens place: $4 \times 7 \times 10 = 280$
There are five 7s in the ones place: $5 \times 7 \times 1 = 35$
total = **86,415**

METHOD 2 *Strategy: Add the numbers.*
Use the standard technique for addition to get 86,415.

FOLLOW UP: Is the sum 86,415 divisible by 15? [Yes]

1B **METHOD 1** *Strategy: Consider the area of 1 small rectangle.*
Since the area of the large rectangle is 70 and the 7 rectangles are congruent, the area of each rectangle is 10. Since the dimensions of each small rectangle are whole numbers, they must be 10 by 1 or 5 by 2. If we try 5 by 2, the dimensions of the larger rectangle would be 10 by 7, and its area would be 70. The perimeter of the large rectangle is $2(10 + 7) =$ **34** cm.

METHOD 2 *Strategy: Consider the dimensions of the larger rectangle.*
The only possible dimensions for the larger rectangle are 2 by 35, 5 by 14, or 7 by 10. The height of the rectangle must be even since it is formed by 2 heights of the small rectangle. Use trial and success to see that, if the height of the large rectangle is 10, the dimensions of the small rectangle must be 5 by 2, so the width of the large rectangle is 7. The perimeter is $2(10 + 7) = 34$ cm.

METHOD 3 *Strategy: Use algebra.*
Let x be the shorter dimension of the small rectangle and let y be the longer one. Then $5x = 2y$ and $(x + y)(2y) = 70$. Divide this second equation by 2 to get $(x + y)(y) = 35$. Since x and y are whole numbers, and since $y < x + y$, $y = 1$ or 5. When $y = 1$, $x = 34$, which makes $x > y$. So, $y = 5$, $x = 2$, and the perimeter is 34 cm.

FOLLOW UPS: (1) How much greater is the sum of the perimeters of the 7 smaller rectangles than the perimeter of the larger rectangle? [64 cm] *(2) How much greater is the sum of the areas of the 7 smaller rectangles than the area of the larger rectangle?* [0 cm^2]

Answer Column
1A **86,415**
1B **34**
1C **10**
1D **37**
1E **40**

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1C **METHOD 1** *Strategy: Use educated guessing.*
Since Beth has 6 more than Connie, guess that Beth has 8, so Connie has 2, Ashley has 16, and Deb has 6. The sum is only 32. Try a larger number for Beth, guess 9. Then Connie has 3, Ashley has 18, and Deb has 9. The sum is only 39. Try a larger number for Beth, guess 10. Then Connie has 4, Ashley has 20, and Deb has 12. The sum is 46. Thus, Beth has **10** stickers.

METHOD 2 *Strategy: Use algebraic reasoning.*
Let A, B, C, and D be the number of stickers held by Ashley, Beth, Connie, and Deb, respectively. Then $A = 2B$, $B = C + 6$, $D = 3C$, and $A + B + C + D = 46$. Replace A, B, and D in the last equation with equivalent expressions containing C. Since $A = 2B = 2(C + 6) = 2C + 12$, we have the equation $(2C + 12) + (C + 6) + C + 3C = 46 \rightarrow 7C + 18 = 46 \rightarrow 7C = 28 \rightarrow C = 4$. Thus, $B = 4 + 6 = 10$.

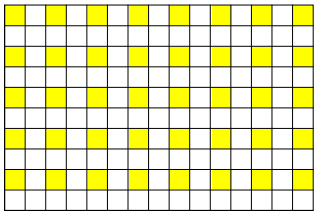
FOLLOW UP: Use the information in the original question except for the total number of stickers. What is the greatest possible total number of stickers the girls can have if it is fewer than 100? [95]

1D **METHOD 1** *Strategy: Observe the pattern and then multiply to calculate future events.*
The first 3-digit number in the list is $92 + 24 = 116$. The next 3 numbers are 140, 164, and 188. The list has four numbers in the 100s. Similarly, the list will have four numbers in each of the 200s, 300s, and 400s. The pattern changes since the number 500 will be one of the 3-digit numbers in the list. Thus, there are five numbers in this group: 500, 524, 548, 572, and 596. The list will have four numbers in each of the 600s, 700s, 800s, and 900s. The last 3-digit number is $1004 – 24 = 980$. The total number of 3-digit numbers is $(8 \times 4) + (1 \times 5) =$ **37**.

METHOD 2 *Strategy: Use division to determine the number of multiples of 24.*
The least 3-digit number in the list is $92 + 24 = 116$ and the greatest is $1004 – 24 = 980$.
Since $980 – 116 = 864$ and $864 \div 24 = 36$, there are 36 jumps from 116 to 980.
The total number of 3-digit numbers in the list is $36 + 1 = 37$.

FOLLOW UP: If the list began with 20, 32, 44, and so on, with each number 12 more than its predecessor, how many 3-digit numbers would be in the list and what is the last number in the list? [75 and 992]

1E *Strategy: Use the grid to recognize how to shade the squares.*
Since there are 15 columns and 10 rows, Melissa would need to shade squares in each of the alternating columns, which would be columns 1, 3, 5, 7, 9, 11, 13, and 15. Shade squares in alternating rows 1, 3, 5, 7, and 9. The maximum result is $8 \times 5 =$ **40** shaded squares. Displayed is one of the possible diagrams.



FOLLOW UP: If the grid size was changed to 15 by 25, what is the greatest number of shaded squares if no two squares share anything including a corner? [104]

NOTE: *Other FOLLOW UP problems related to some of the above can be found in our five contest problem books and in "Creative Problem Solving in School Mathematics." Visit www.moems.org for details and to order.*